medAIval: Automated ICD-10 Coding from Clinical Notes

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Introduction

- Manual ICD-10 coding from clinical notes is a labor-intensive process prone to human error [1].
- Accurate coding is essential for accurate hospital billing and patient record consistency.
- Diagnoses in the J01 (acute sinusitis) and J32 (chronic sinusitis) families present a unique challenge due to their semantic similarity and extremely nuanced text descriptions.
- The inherent variability and unstructured nature of clinical notes along with their lack of complete and specific labeling presents a unique challenge.
- Enhancing automation in ICD-10 coding could significantly reduce administrative burdens in healthcare and make patient data more useable for other ML tasks
- Other existing works are generally trained on datasets like MIMIC, taking advantage of an easily distinguishable set of ICD codes.

Data, Preprocessing, and Features

- IRB-approved access to Stanford STARR database; extracted approximately 4,400 patients at the Stanford ENT Clinic.
- 9GB of data comprising clinical notes, pathology reports, radiology results, lab orders, prescriptions, and diagnosis codes.
- HIPAA compliance ensured using Stanford's in-house STRIDE NLP model, which scrubs PHI data while preserving unique patient IDs and patient-relative dates.
- Pipeline 1 (classic text cleaning): whitespace removal, lowercasing, stopword elimination, tokenization with BioClinicalBERT
- Pipeline 2 ("synonym" based approach): UMLS-based mapping[8] to generate sparse "synonym" vectors from 7,685 referenced codes.
- Our input is the result from Pipeline i and the output is a binary 21-vector of diagnoses with the order matching the set of classes in Figure 1

class_imbalance_ratio.png

Figure 1. Ratio of classes relative to the median to highlight diagnostic imbalances.

cocurtop8.png

Figure 2. Co-occurence of top 8 diagnosed codes in our set

top3_co_occurrence.png

Figure 3. Co-occurence of top 3 diagnosed codes in our set

Binary Classifiers

- Individual binary classifiers for each given ICD-10 code using data from Pipeline 1
- Input token is converted into an embedding (1)
- Using pretrained weights from the Bio_ClinicalBERT word encoder, the embedding is passed through 12 transformer layers, consisting of redmulti-headed attention and a feed-forward network.
- We use cross-entropy loss on the two output logits where $\hat{p}_c$ is the predicted probability for the correct class (2)
- Our ground truth label is represented by $y \in \{0, 1\}$.

Two Towers: LeBron Style

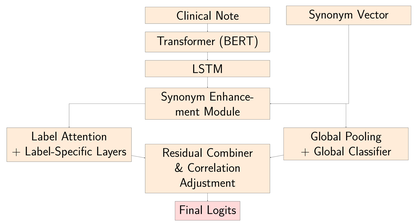

Figure 4. Simplification of the LeBron Two Tower model architecture. It's named after LeBron; we love him!

ablation_study_poster.png

Figure 5. Resulting F1 scores of eliminating specific layers/paths (i.e. one tower at a time)

- Idea: combine the unique features from *both* clinical text and medical synonyms. Dense text embeddings with sparse synonym vectors together improve prediction of hard-to-identify ICD-10 codes.
- Text Tower: Clinical text embeddings from Bio_ClinicalBERT are processed through a bidirectional LSTM, capturing sequential context. Uses a label-specific attention mechanism (LeBronWiseAttention) to focus on key text segments for each ICD-10 code.
- Synonym Tower: SynoBronEnhancementModule intertwines cross-attention with a separate Vector Encoder model allowing text features to dynamically attend to synonym features, effectively merging unstructured text and structured synonym information.
- The Junction: Combines label-specific and global predictions using a learnable residual weight parameter, while incorporating label correlation patterns for better prediction coherence.
- Loss: Addresses class imbalance in ICD-10 codes through a weighted combination of Focal Loss (LeFocalYouLoss) and Asymmetrical Loss (AsymmetricalLeBronLeLoss).

$$\text{attn_scores} = \frac{\text{Query(text_feat)} \cdot \text{Key(syn_feat)}^T}{\sqrt{d}} \tag{3}$$

$$\text{logits} = \alpha \cdot \text{label_specific_logits} + (1 - \alpha) \cdot \text{global_logits} + \text{correlation_adjustment} \tag{4}$$

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{focal}} + 0.5 \cdot \mathcal{L}_{\text{asymmetric}} \tag{5}$$

Training Results

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Final Poster/distribution.png

Discussion and Analysis

- We compare the test F1, accuracy, and recall of our final models with prior works
- Our classification task is more challenging due to a messy training data and niche ICD-10 with high inter-code semantic similarity, unlike the broader, cleaner, less overlapping MIMIC dataset used in previous studies.

Table 1. Micro-F1 Scores for Existing Works [9]

Model Name	Micro-F1
MSMN	.758
PLM-ICD	.755
CAML	.692

Table 2. Performance Metrics of Our Models

Model Name	F1 Score	Accuracy	Recall
Binary Classifiers	0.792 (avg)	0.779 (avg)	0.857 (avg)
Vector Encoder	0.663	0.692	0.538
Two Towers	0.512	0.936	0.683

- Because our binary classifier pipeline trains a seperate model for each ICD-10 code, the aggregated model can maintain ICD-10 code specific, which translates to more flexibility and better performance at the cost of higher compute costs.
- Our LeBron Two Tower LSTM-and-encoder based approach also performs better than some of our benchmarks, which is likely because of nested attention mechanisms that learn to emphasize certain input.

Conclusion and Future Work

- Our Two Tower model achieved micro-averaged F1 scores of approximately 0.51, which is significant given the semantic similarity between target codes and the extreme rarity of some conditions.
- We also proposed a binary classifiers method which had the best F1 score of 0.792 (as an average over all of them) but a worse accuracy of 0.779.
- We did expect the combined model to perform better than any of its children components. We didn't expect the averaged binary classifier model to be so powerful!
- We observed that our models achieved better performance when they assigned custom weights to each ICD-10 code or used nested attention to focus on key segments, compared to generalized models that inferred codes from the entire input.
- Future research could extend this work by incorporating larger, cleaner datasets, implementing hierarchical classification to leverage parent-child code relationships, integrating advanced language models, and incorporating temporal modeling to better distinguish between acute and chronic conditions.

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